

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

MCA101: Discrete Structures (New Course)

MCA111: Discrete Mathematical Structures (Old Course)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. What are algebraic structures? Let $G=\{1, -1, i, -i\}$ be a set. Prove that $(G, *)$ is a group under the binary operation $*$ which implies multiplication. 4+8
2. What are Boolean functions? Explain SOP and POS. Simplify Boolean function using K-map in SOP and POS forms and implement with gates. 2+4+6

$$F(A, B, C) = \sum m(0, 2, 3, 4, 5, 6)$$

- 3(a) What is recurrence relation? Solve the following recurrence relation: $a_n=2 a_{n-1}, n \geq 1$ and $a_0=3$ 6

- (b) Using mathematical induction, show that: 6

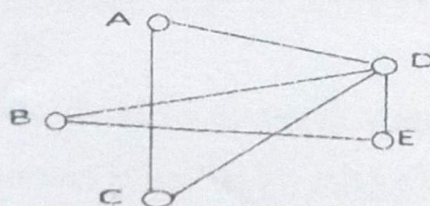
$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Group B

Answer SEVEN questions.

7×8=56

4. Use Fleury's algorithm to find an Euler circuit for the graph given below starting at vertex A.

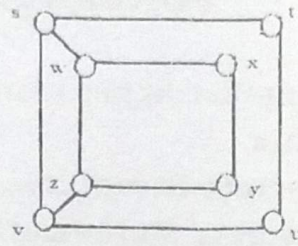
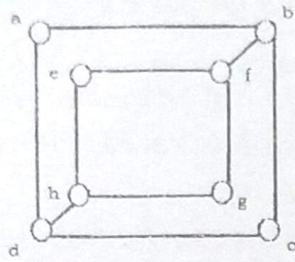


5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^2 + 1$. Find $f^{-1}(-8)$ and $f^{-1}(5)$.

Contd. ...

(2)

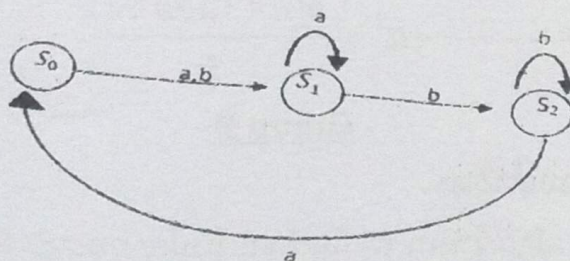
6. Determine whether the graphs shown below are isomorphic.



7. What do you understand by permutation? Find the total number of permutations of the word 'STATISTICS'.
8. What do you mean by even, odd and cyclic permutation functions? Determine whether the following permutation is even or odd.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 4 & 3 & 5 & 2 \end{pmatrix}$$

9. If R is a relation in the set of integers Z defined by, $R = \{(x, y) : x \in Z, y \in Z, (x-y) \text{ is divisible by } 6\}$. Prove that R is an equivalence relation.
10. Write converse, inverse and contrapositive of the followings:
- (a) All glitters is not gold.
 - (b) All hexagon are regular polygon.
 - (c) Fruit juice contain vitamin C.
11. Construct the state transition table of the finite state machine whose diagram is shown below.



12. Write short note on (Any TWO):
- (a) Network flows
 - (b) Homomorphism
 - (c) Tautology